Lab - Ohms Law

Answer the following questions based on electricity and Ohm's Law. Show all steps when solving problems.

a. What are the four basic units of electricity? Provide the variable name and symbol, and unit name and symbol.

The four basic units of electricity are: Voltage (V), measured in Volts (V); Current (I), measured in Amperes (A); Resistance (R), measured in Ohms (Ω); and Power (P), measured in Watts (W).

b. Write the equation for Ohm's Law.

The equation of Ohm's Law is V=IR, where V is voltage equal to I, the current, multiplied by R, the resistance.

c. Re-arrange the Ohm's Law equation to solve the following:

I = V/R

R = V/I

d. Power is equal to voltage multiplied by current. Add the missing information in each of the following power equations.

P = V*I

$P = R*I^2$

$P = V^2/R$

e. The yellow wire connected to a power supply carries 12V. If the power supply provides 60W of power to the yellow wire, how much current is passing through the yellow wire?

Voltage = 12V

Power = 60W

Current =?

I = P/V

I = 60W/12V

I= 5A

There are 5A of current passing through the yellow wire.

f. There are 3.3V passing through an orange power supply cable, and there are 0.25 ohms of resistance in the orange wire. How much power is supplied to the orange wire by the power supply?

Voltage = 3.3V

Resistance = 0.25 ohms

Power = ?

$P = V^2/R$

$P = (3.3V)^2/0.25Ohms$

P = 10.89V/0.25Ohms

P= 43.56W

There are 43.56W of power supplied in the orange wire.

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g. A wire from the power supply is carrying 120W of power and 24A of current. How much power is supplied to the wire by the power supply?

Power = 120W

Current = 24A

Voltage =?

V=P/I

V= 120W/24A

V = 5V

There are 5V of power is supplied to the wire by the power supply.

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